Plasma proteins

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Critical Functions of Plasma Proteins

Plasma proteins perform numerous essential functions that maintain homeostasis and protect our health. Understanding these functions reveals why monitoring protein levels is so important in clinical practice.

Oncotic Pressure Maintenance

Albumin accounts for 40-60% of blood proteins, making it a major contributor to maintaining the oncotic pressure of plasma and preventing fluid loss to the tissues.

Substance Transport

Transporting water-insoluble substances, such as fatty acids, steroid hormones, vitamins, calcium, and bilirubin throughout the body.

Acid-Base Balance

Maintenance of acid-base balance, ensuring proper pH levels are maintained in blood and tissues.

Amino Acid Reserve

Circulating reserve of amino acids for cells, especially albumin, providing building blocks when needed.

Hemostasis

Numerous factors related to coagulation and fibrinolysis are found in plasma, essential for blood clotting.

Defense Mechanisms

Immunoglobulins and C-reactive protein are involved in protecting the body against pathogens and infections.

Analysis of plasma proteins:

1. *Plasma Total protein concentration*
2. *Serum protein electrophoresis or proteinogram*
3. *Measurement of specific proteins*
4. *Immunofixation*

1. Plasma total protein

Plasma or serum

Measurement ALL of the proteins in plasma

6.6-8.7 g/dL

HIPOPROTEINEMIA

More frequent than hyperproteinemia

Causes:

1 Decreased intake (abnormal diet, malabsorption…)
2 Failure in synthesis (hepatic disease, severe immunodeficiency
3 Hemodilution (fluid retention
4 Increased losses (hypermetabolic states, kidney disease, extensive burns..)

HIPERPROTEINEMIA

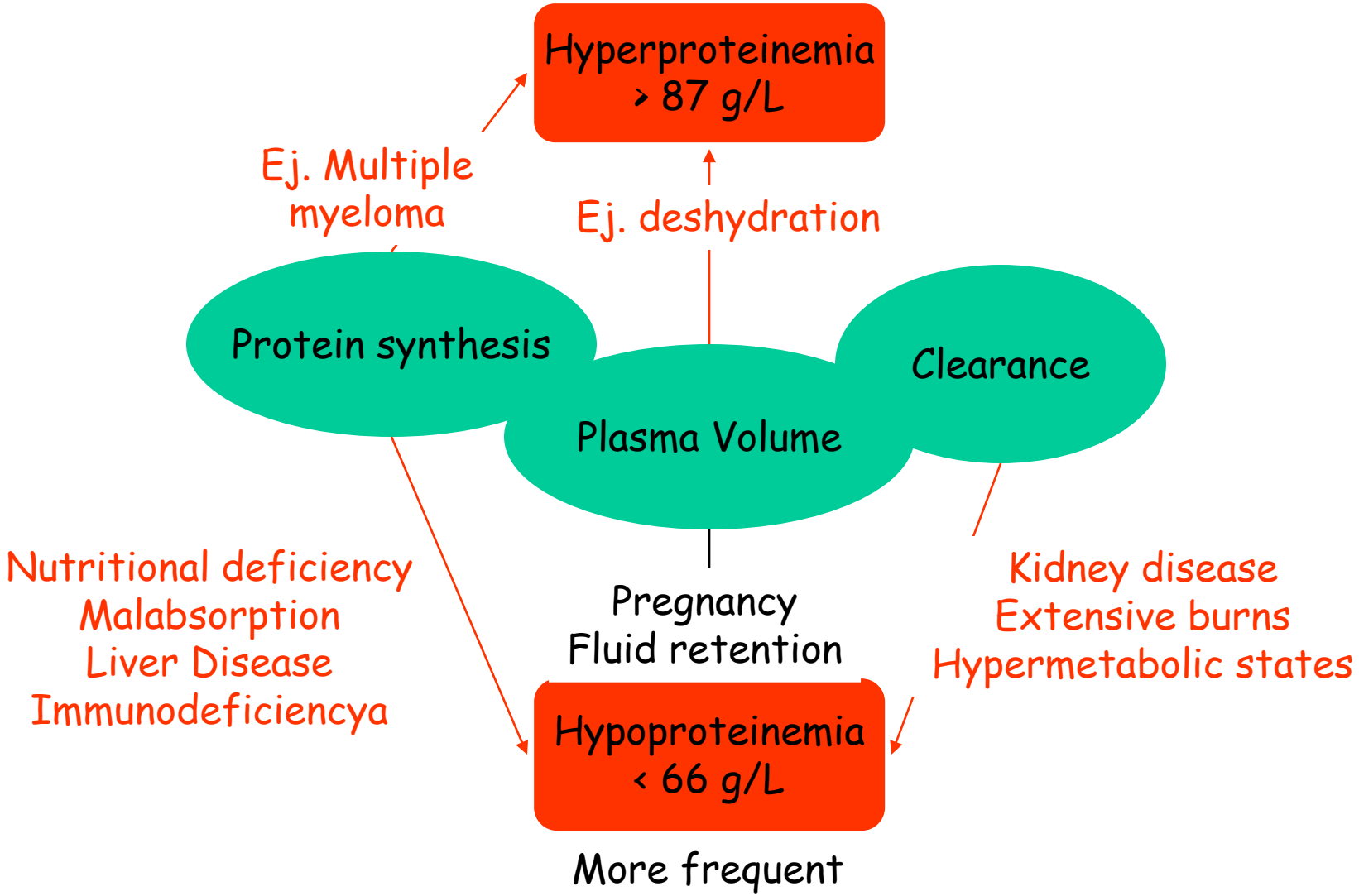
Causes:

1. impaired synthesis
    1. Multiple myeloma. Increased monoclonal synthesis of immunoglobulins
    2. Polyclonal increases of immunoglobulins

2. hemoconcentration.

dehydration may be due to decreased fluid intake or increased fluid loss, as in the case of severe diarrhea or vomiting.

Changes inTotal Protein concentration in blood



2. Serum protein electrophoresis

- Separated by electrical charge (depends on the pH of the buffer)
  - agarose gel. stained using Coomassie brilliant blue
  - capillary. absorbance of the peptide bond with UV light.

- Total protein concentration is required to quantify each fraction.

- Immunoglobulins migrate in the opposite direction due to the low charge in relation to electroendosmosis (movement of cations towards the cathode which pulls the surrounding water molecules).

2. Serum protein electrophoresis

01

Sample Preparation

Serum sample is prepared and applied to agarose gel or capillary system

02

Electrophoresis

Proteins migrate based on electrical charge at pH 8.6

03

Staining or Detection

Coomassie brilliant blue staining for gel or UV absorbance for capillary

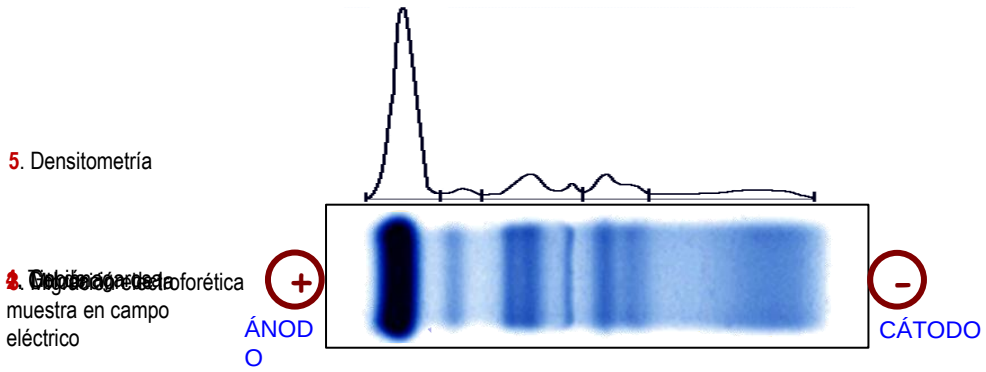
04

Analysis

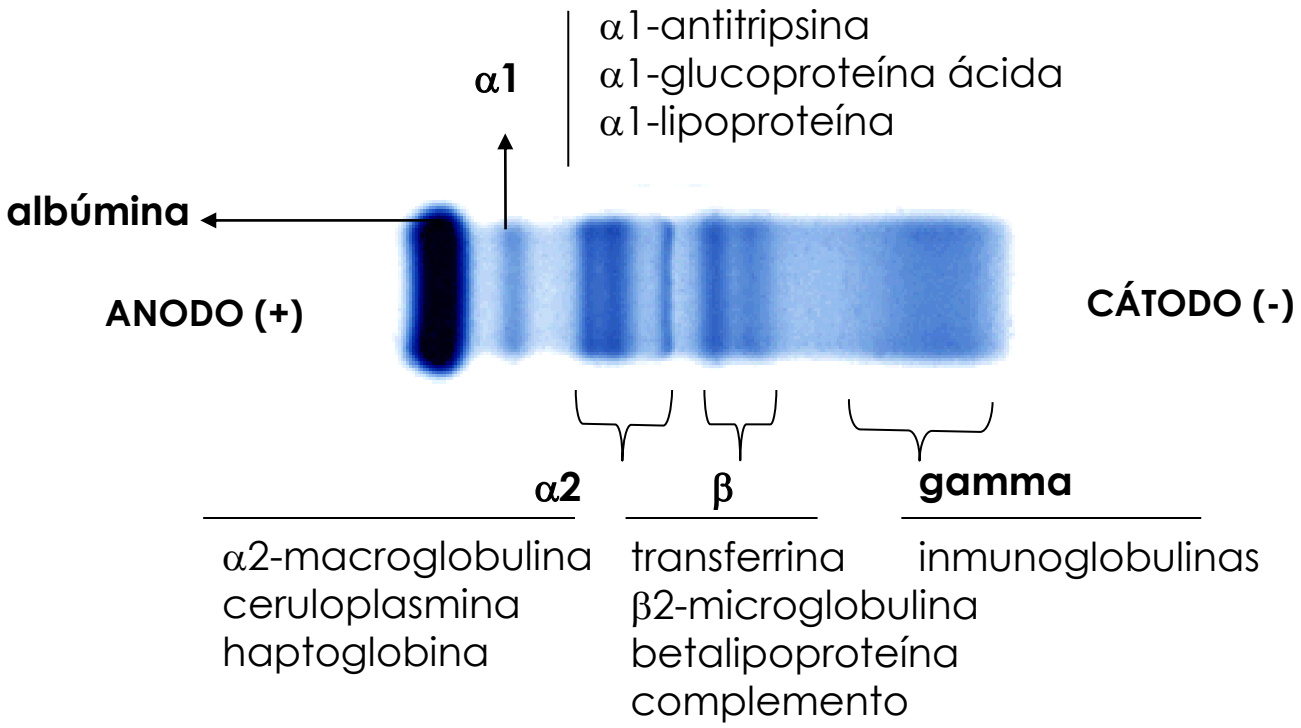
Visual inspection and densitometry measurement to obtain proteinogram profile

Electrophoresis separates the proteins into **five main bands or fractions**: albumin is at the anodic (+) end, followed by the α1-globulin, α2-globulin, β-globulin and γ-globulin fractions. The proteinogram should be performed with a serum sample, since with plasma, fibrinogen appears as an additional band in the β-region, similar in appearance to a monoclonal band, which can lead to an error in interpretation.

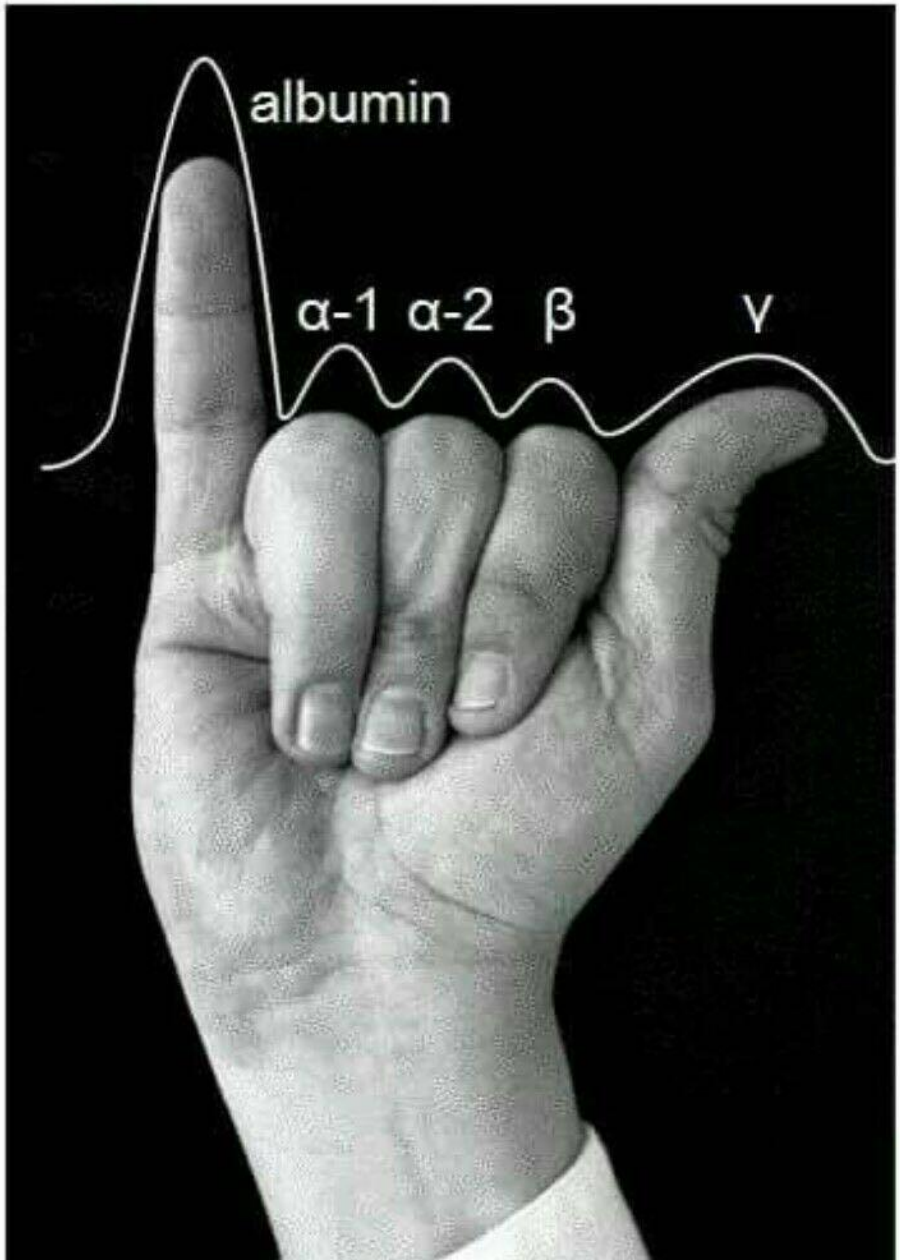
Proteinograma



% cada banda y PT → Concentración de proteína en cada banda


α1-antitripsina
α1-glucoproteína ácida
α1-lipoproteína
α1
albúmina
ANODO (+)
CÁTODO (-)
α2
β
gamma
α2-macroglobulina
ceruloplasmina
haptoglobina
transferrina
β2-microglobulina
betalipoproteína
complemento
inmunoglobulinas

Easy way to remember serum protein electrophoresis



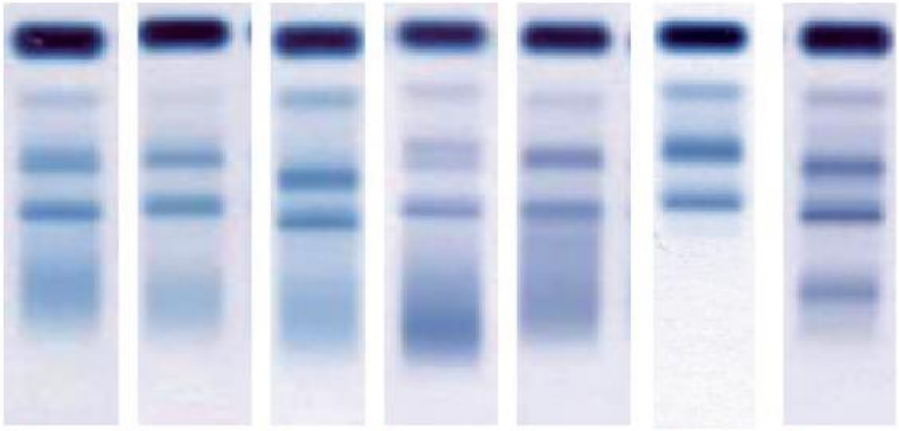
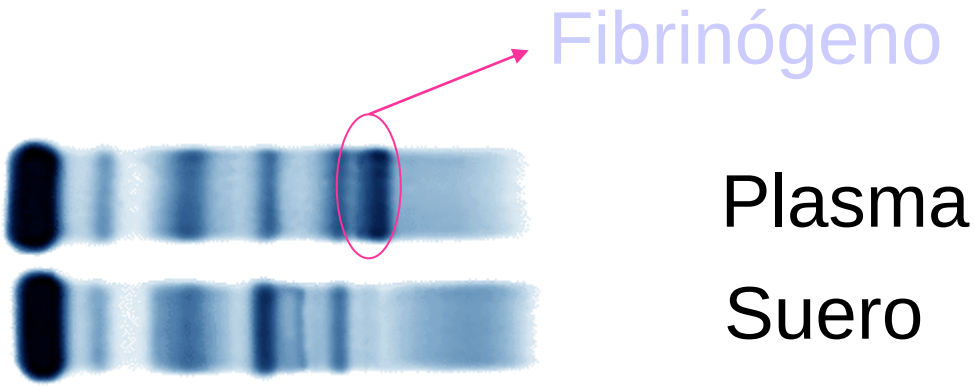


Figura 16-3 Ejemplos de alteraciones del proteinograma. **A.** Normal. **B.** Ausencia de banda en $\alpha 1$ por deficiencia de $\alpha 1$-antitripsina. **C.** Incremento de $\alpha 1$, $\alpha 2$ y descenso de β en una reacción de fase aguda. **D.** Incremento de la región γ en una infección. **E.** Incremento de la región γ con puente β-γ en una infección hepática. **F.** Notable disminución de la región γ en una inmunodeficiencia. **G.** Presencia de una banda monoclonal en la región γ.

LA MUESTRA DEBE SER SUERO



3. Measurement of specific proteins

Albumin: The Most Abundant Plasma Protein

Albumin is the most abundant protein in plasma, making it the main protein responsible for maintaining oncotic pressure. It has a large amount of negative charges in its molecule.

40-60%

Percentage of total plasma proteins

68.5

Molecular Weight

kDa

21

Half-Life

Days in circulation

Key Functions:

- Maintains oncotic pressure, preventing fluid loss to tissues
- Serves as reservoir of amino acids for cellular use
- Transports fatty acids, drugs, vitamins, calcium, bilirubin, and various hormones

Albumin Synthesis and Metabolism

Synthesized in the liver according to protein intake and its concentration in blood. Mmetabolized in many tissues where it is taken up and hydrolyzed by the cells, taking advantage of its amino acids. Catabolism increases in traumatic processes, infections and surgical interventions. Changes in the plasma concentration of albumin modify the concentration of the substances it transports. frequently used in monitoring the patient's nutritional status

Clinical significance of Albumin changes

Hypoalbuminemia (much more common)

Lower Synthesis

Malnutrition with low amino acid intake, intestinal malabsorption, or liver diseases with decreased synthetic capacity

Hemodilution

During pregnancy, circulating volume increases causing hemodilution and decreased albumin concentration

Excessive Losses

Renal disorders can cause very severe hypoalbuminemia through high protein losses via the kidney

Hyperalbuminemia

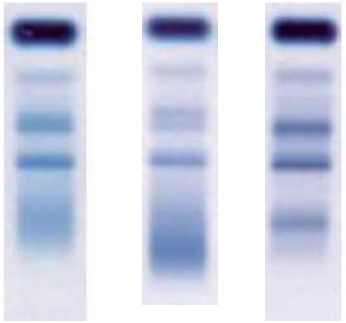
Deshidratación

Artificially

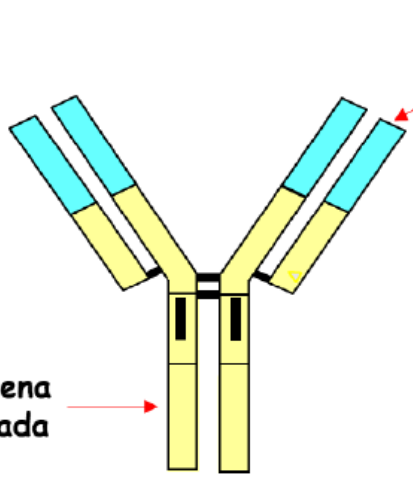
Prolongued tourniquete

Immunoglobulins

- Synthesized by plasma cells derived from B lymphocytes
- Migrate to the gamma región in the electrophoresis
- Function: antibodies



Structure



G, A, M, D, E

5 types (IgG, IgA, IgM, IgD and IgE)

Clinical significance

1. Decrease. Hypogammaglobulinemia

Immunodeficiency: fallo en la síntesis o pérdidas

2. Increase. Hypergammaglobulinemia

- polyclonal: Difuse band because all immnoglobulins increase infection, inflammatory processes.

- monoclonal: one clon, unique band

monoclonal gammopathies

multiple myeloma, etc

Multiple Myeloma: A Monoclonal Gammopathy

Multiple myeloma is a malignant neoplasm characterized by the elevation of a clone of plasma cells in the bone marrow, which produce a single type of immunoglobulin, making it monoclonal. At the same time the production of normal immunoglobulins is inhibited, so that these patients frequently present recurrent infections.

Osteolytic Lesions

Bone destruction with elevated alkaline phosphatase and calcium in blood

Monoclonal Protein

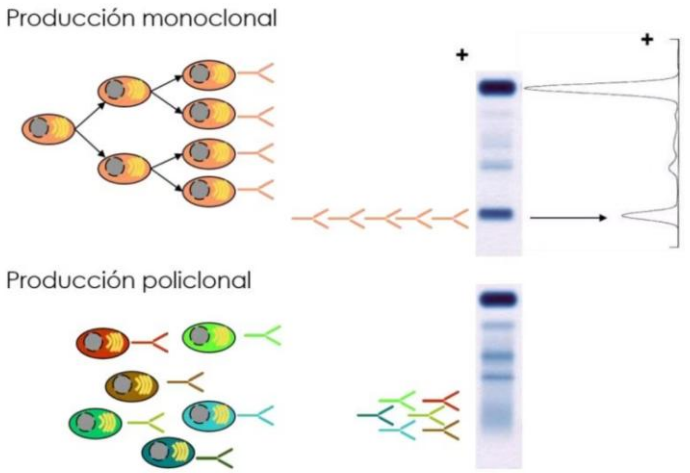
Single clone of plasma cells producing identical immunoglobulins

Immunosuppression

Inhibited production of normal immunoglobulins leading to recurrent infections

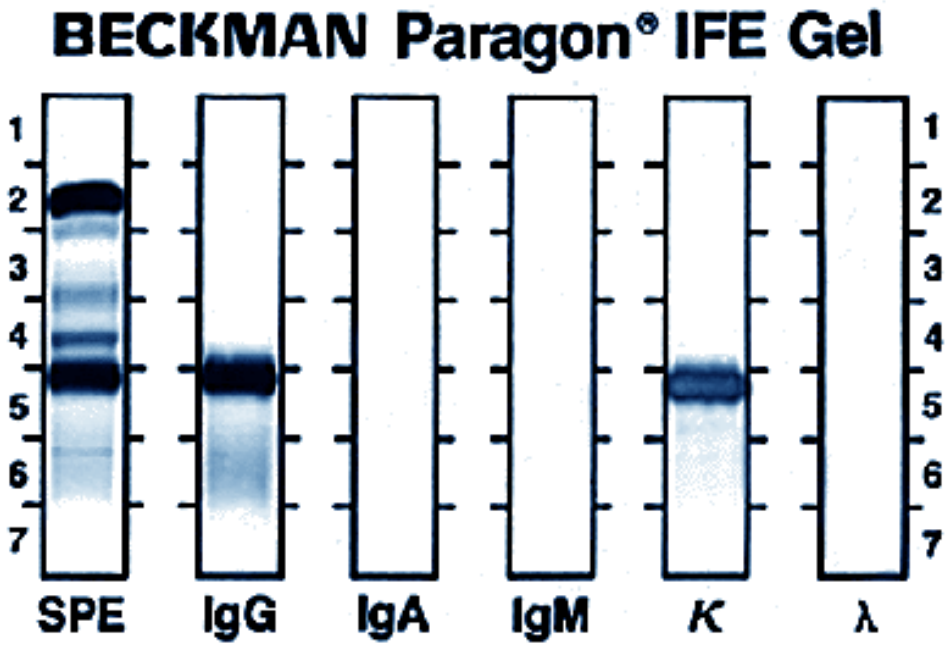
Systemic Symptoms

Weight loss, anemia, hemorrhages, or renal failure



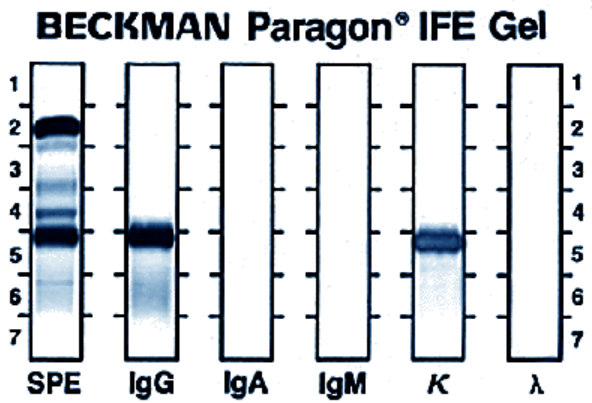
4. Immunofixation

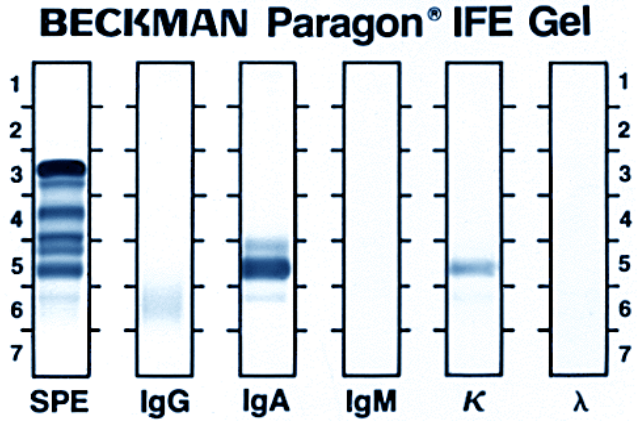
INMUNOFIJACIÓN: PROCEDIMIENTO

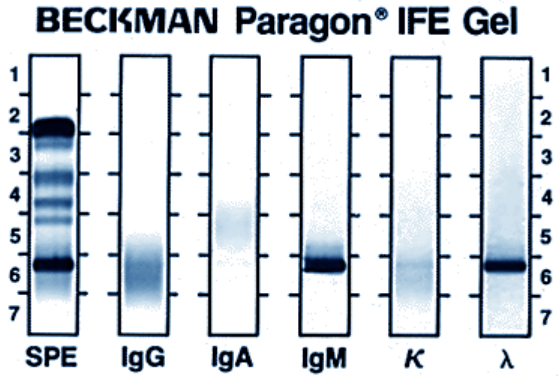


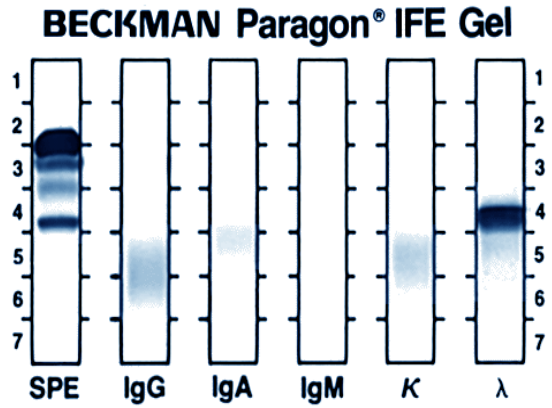
- Aim: Identify the isotype of a monoclonal protein
- Steps:
  - 1.- Agarose gel electrophoresis to separate proteins
  - 2.- immunoprecipitation with specific antibodies against heavy and light chains
  - 3.- Staining
- More sensitive than serum protein electrophoresis

inmunofijaciones de sueros con bm







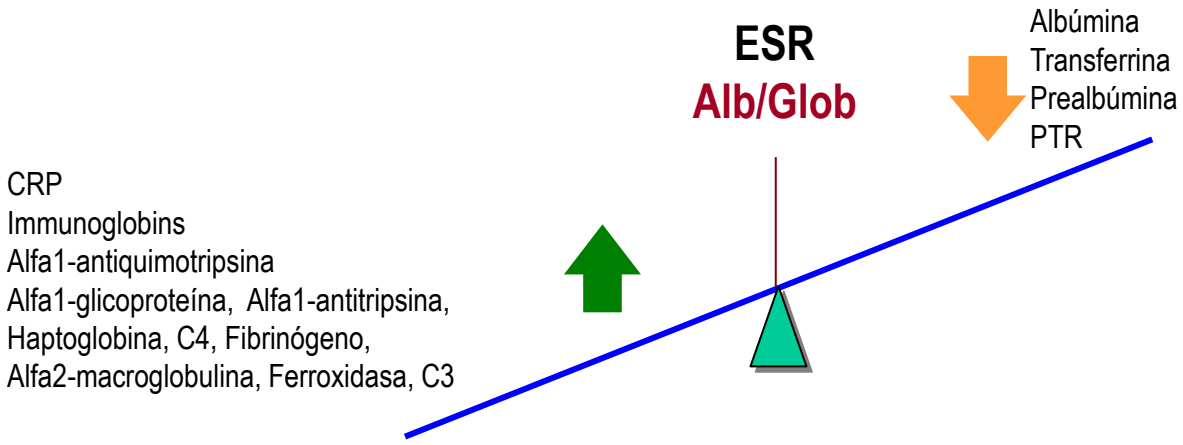


ACUTE PHASE REACTION

Acute Phase Reaction

Systemic systemic reaction against infection, inflammation or tissue injury (mechanical, physical, chemical, biological or tumor). Certain proteins change increase their concentration in the blood and other decrease. The changes are not disease-specific.

Positive Acute Phase Reactants (Increase)	Negative Acute Phase Reactants (Decrease)
Proteins like C-reactive protein (CRP) immunoglobulins	Albumin

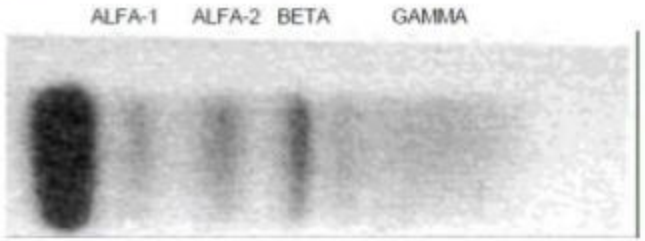


Erythrocyte Sedimentation rate

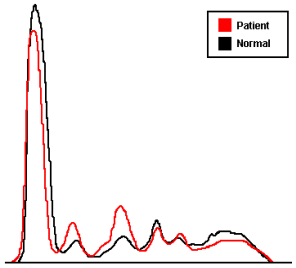
The ratio between albumin and globulins changes, leading to increased red cell sedimentation rate due to altered plasma properties

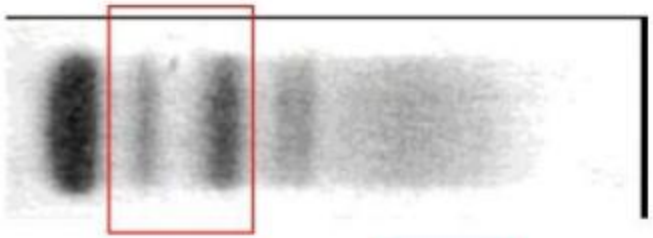


Reacción Inflamatoria

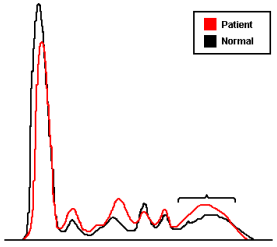


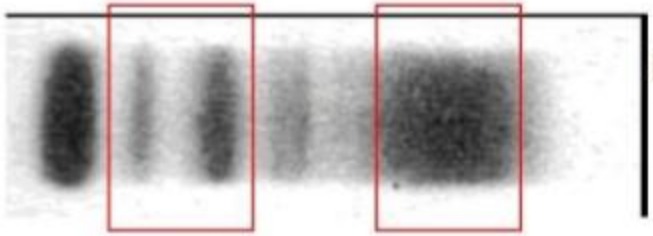
NORMAL





REACCION INFLAMATORIA
(FASE AGUDA)





REACCION INFLAMATORIA
(CRONICA)

Patrón similar al de inflamación aguda + elevation of gamma

C-Reactive Protein

Synthesized in the liver.

Its concentration increases dramatically soon after cell injury. Its elevation is not disease specific

Clinical Applications:

- Detect inflammation, including chronic inflammation in autoimmune diseases
- Very sensitive for diagnosing inflammatory intestinal diseases
- Serial determination reflects severity and evolution of the process
- Monitoring treatment efficacy